Sign Language Detector

How to run the project on another Windows PC

The repository already contains the trained models, so predictions work straight after cloning. You do not need the dataset unless you want to retrain. The app recognises 20 ASL words: **yes, no, eat, drink, like, want, help, need, go, what, who, mother, woman, man, finish, computer, family, fine, school, work**.

1. Install the prerequisites (one time)

Tool	Version	Check with
Git	any recent version	git --version
Python	**3.10, 3.11 or 3.12** (3.11 recommended). **Not 3.13 or newer**: MediaPipe has no build for it.	python --version
Node.js	20 or newer (LTS)	node --version
Browser	Chrome or Edge	
Webcam	only needed for the Live webcam tab	

About 3 GB of free disk space is needed (PyTorch is large). While installing Python, tick **Add python.exe to PATH**.

2. Download the project

Open PowerShell where you want the project folder, then run:

```
git clone https://github.com/saravanan2736/sign-language-detector.git
cd sign-language-detector
```

3. Start everything with one command

Double-click **run.bat** in the project folder, or run this in PowerShell:

```
.\run.ps1
```

On the first run it creates the Python environment, installs the Python and web packages (this takes a few minutes), starts the backend and the web UI, waits until both respond, and opens **http://localhost:3000** in your browser. Press **Q** or **Ctrl+C** in that window to stop both.

> If PowerShell says scripts are disabled, run this once and try again: `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`. Double-clicking run.bat does not need this.

Command	What it does
.\run.ps1	start both, open the browser, stop with Q or Ctrl+C
.\run.ps1 -Detach	start both in the background and return
.\run.ps1 -Stop	stop servers started with -Detach
.\run.ps1 -NoBrowser	do not open the browser

4. Using the app

Tab	How to use it
Upload video	Click a sample clip (hover to preview it) or drop your own video of one sign, then read the prediction. The panel shows the best guess, a confidence score, the top 5 guesses and an animated replay of the hand and body landmarks the model saw.
Live webcam	Click **Start camera** and allow camera access. Press **Start recording** (or the Space key), wait for the 3-2-1 countdown, make one sign. Recording stops when your hands leave the picture.
History	Predictions made from uploaded videos.

The first prediction after starting is slow (up to about 20 seconds) while MediaPipe warms up. Do one test prediction before showing it to anyone.

5. Manual setup (alternative to step 3)

Use this if you prefer to run each part yourself. Open **two** PowerShell windows in the project folder.

One time: create the Python environment and install packages

```
python -m venv venv
.\venv\Scripts\Activate.ps1
pip install -r requirements.txt
```

pip may warn that streamlit wants a newer protobuf. That is expected and the app still works.

Window 1: backend (wait for `Application startup complete`; http://localhost:8000 should show a running message)

```
.\venv\Scripts\Activate.ps1
cd backend
python main.py
```

Window 2: web UI (then open http://localhost:3000)

```
cd web
npm install
npm run dev
```

6. Optional extras

What	Command (project folder, venv activated, backend running where needed)
Streamlit version of the UI	cd frontend then streamlit run app.py (opens at http://localhost:8501)
Native webcam window	python live_camera.py (press Q to quit)
Run the tests	pytest tests

7. Troubleshooting

Problem	Fix
pip install fails on mediapipe	You are on Python 3.13 or newer. Install Python 3.11 and delete the venv folder, then run again.
"module mediapipe has no attribute solutions"	A newer MediaPipe was installed. Run: pip install mediapipe==0.10.14 "protobuf>=4.25.3,<5"
Web UI shows "Backend offline"	The backend is not running or is on another port. Start it (step 3 or 5).
Port 8000 or 3000 already in use	Close the program using it, or run .\run.ps1 -Stop.
Camera does not start	Allow camera access in the browser. The camera only works on http://localhost or HTTPS, not on a plain LAN IP address.
Predictions are very slow	Another program is using the CPU. Close it. MediaPipe runs on the CPU.
npm install fails	Check node --version is 20 or newer. Delete web\node_modules and retry.
Looks stale after a change	Stop the servers, delete the web\.next folder, and start again.
Where are the logs?	In the logs folder (backend.log, web.log and the matching .err.log files).

What is not included. The training videos and cached landmarks (data/raw, data/processed) are not in the repository because of their size. Running the app works without them. Retraining and evaluate.py need the WLASL-100 dataset in the data folder. Accuracy is modest (20-word model: about 26% top-1 and 61% top-5 on unseen test clips), because the dataset has only about 7 clips per word.